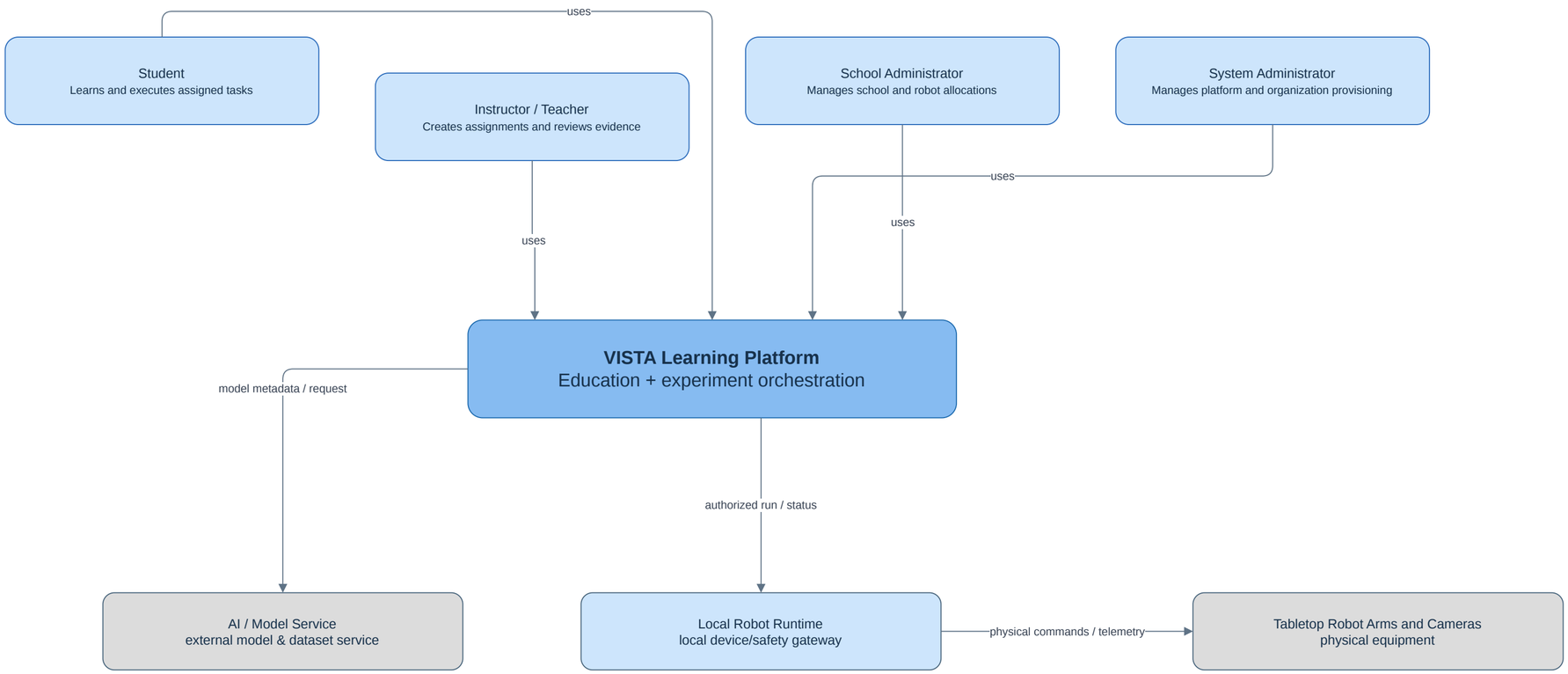
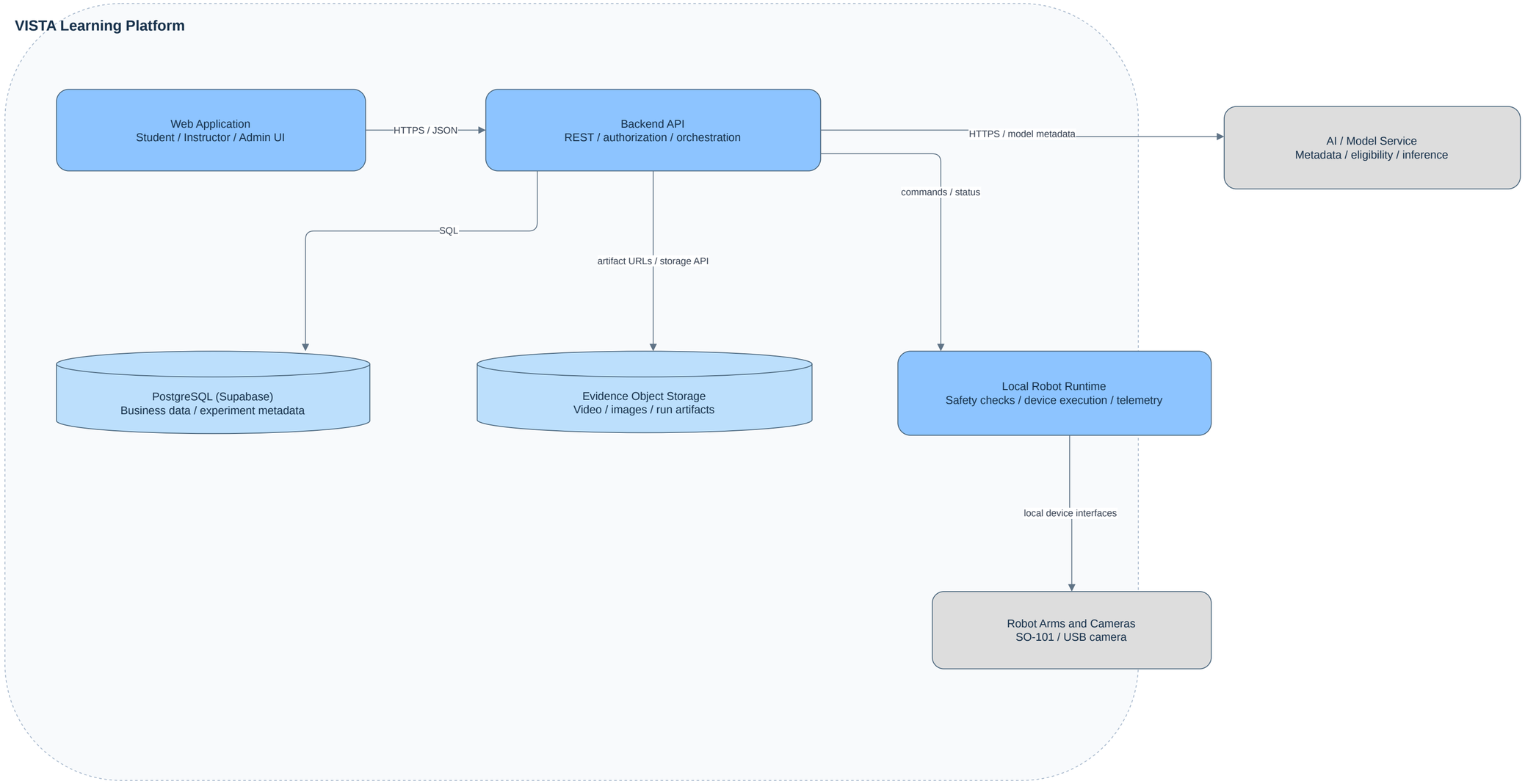
VISTA - C4 System Context

Reviewed conceptual redraw from provided screenshot; architecture must be checked against source code



VISTA - C4 Container Diagram

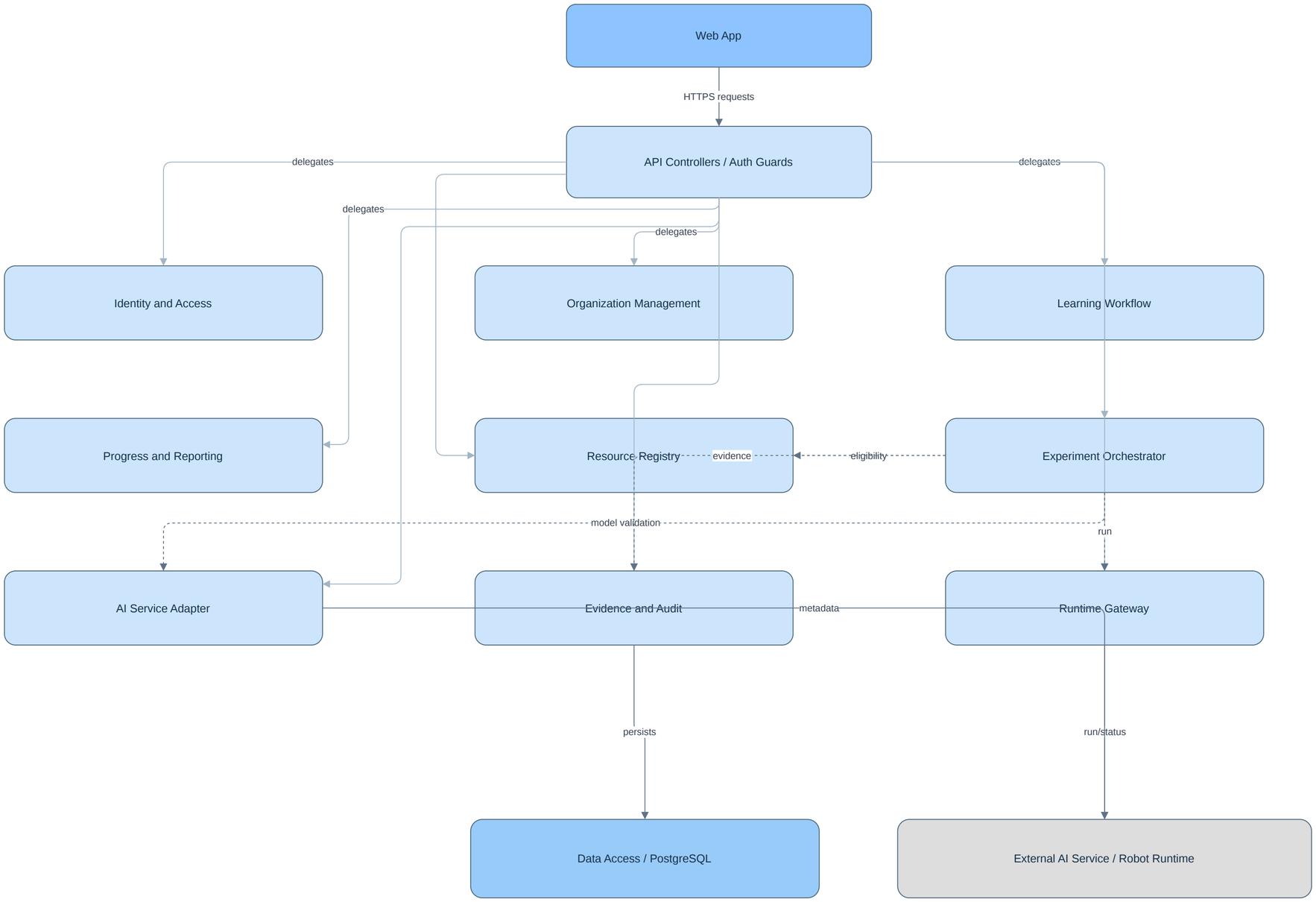
Fixed duplicate Backend API and mislabeled PostgreSQL; deployment boundary is conceptual



Boundary decision: verify whether Local Robot Runtime is deployed inside or outside the VISTA system boundary. Distinguish intended architecture from currently running components.

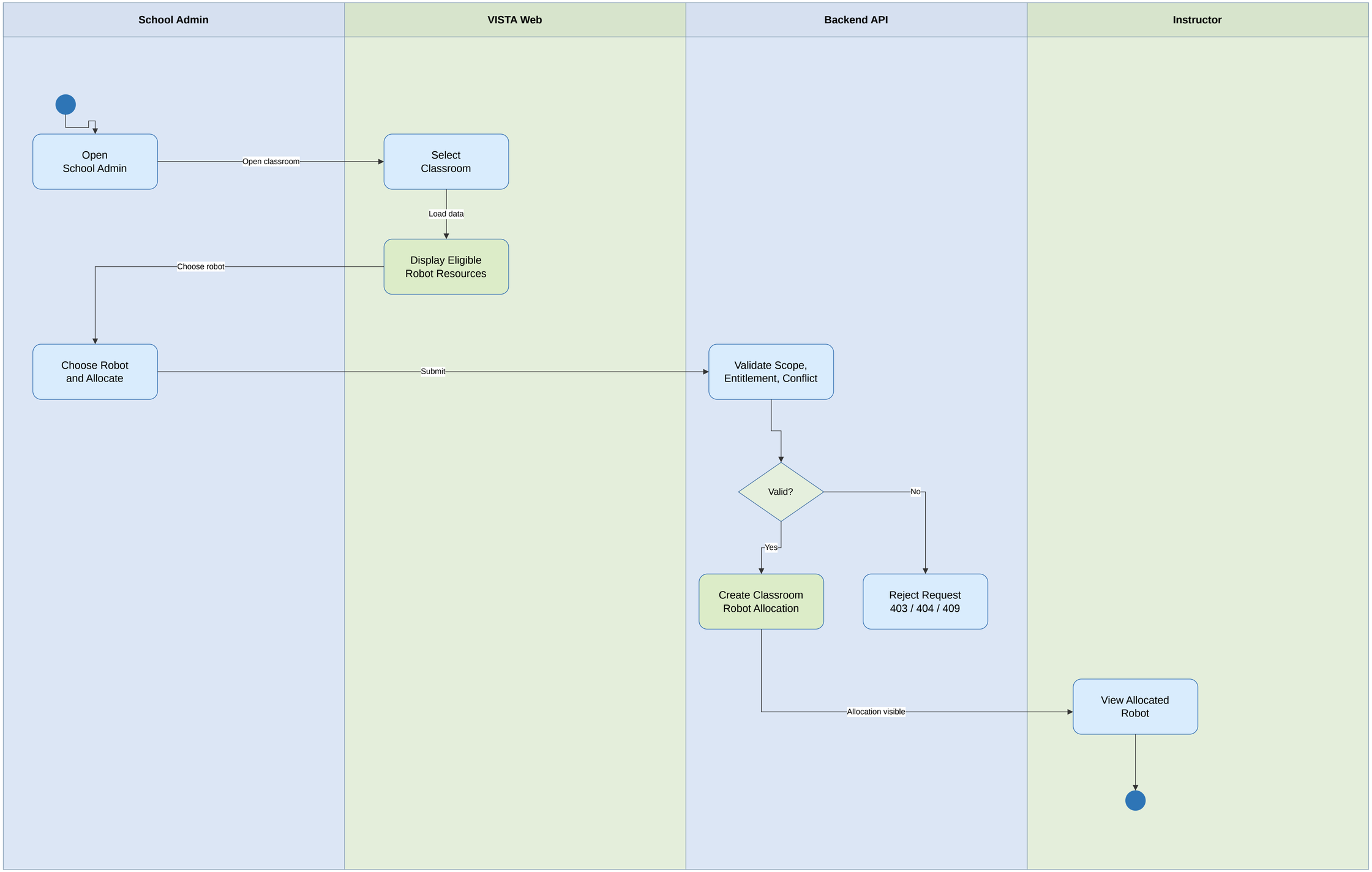
VISTA - Backend API Components

Conceptual component view rebuilt from provided screenshot; verify module names against current NestJS code



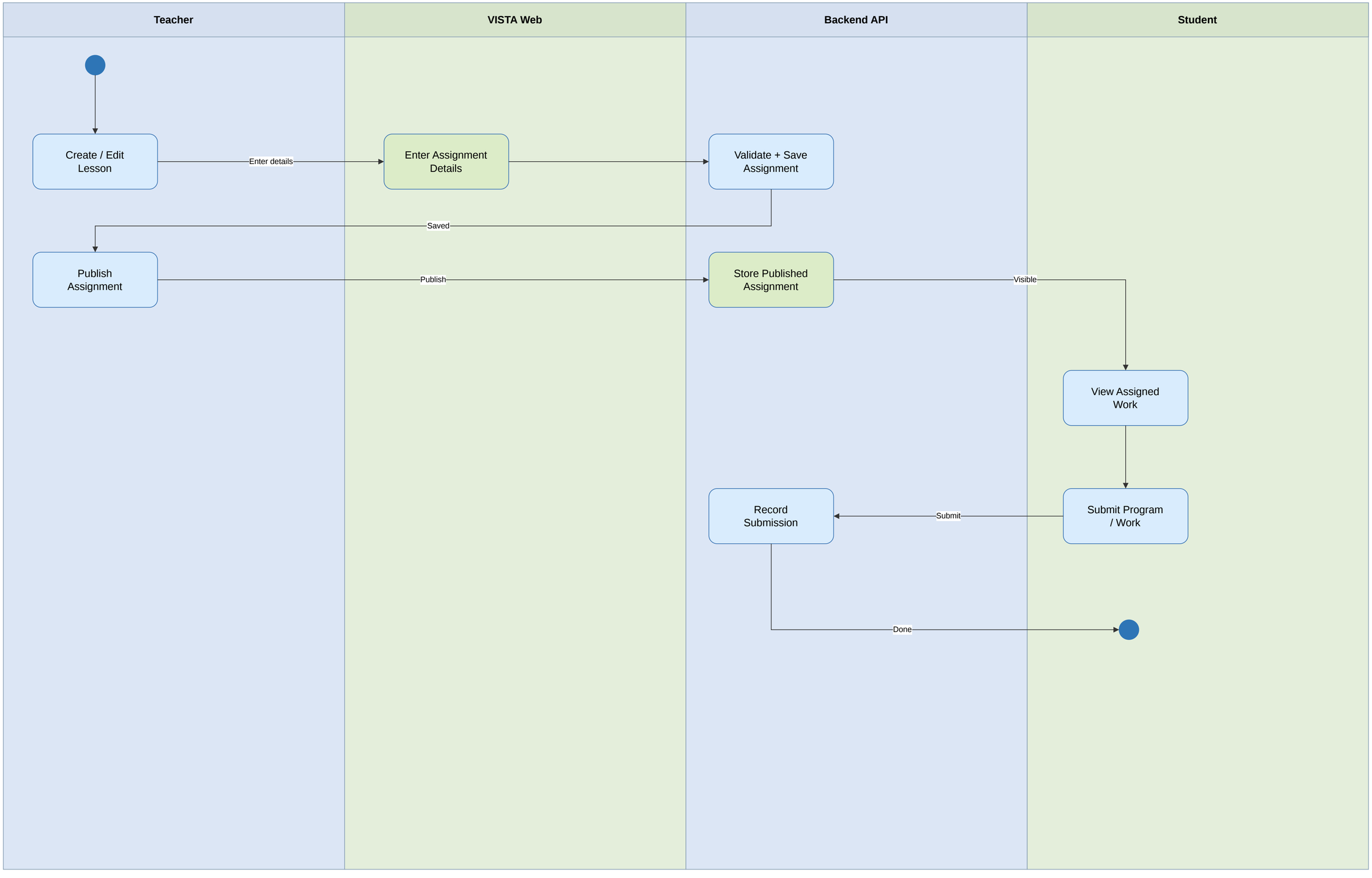
04 Activity WF-01 - School Administration & Robot Resource Allocation

Activity Diagram



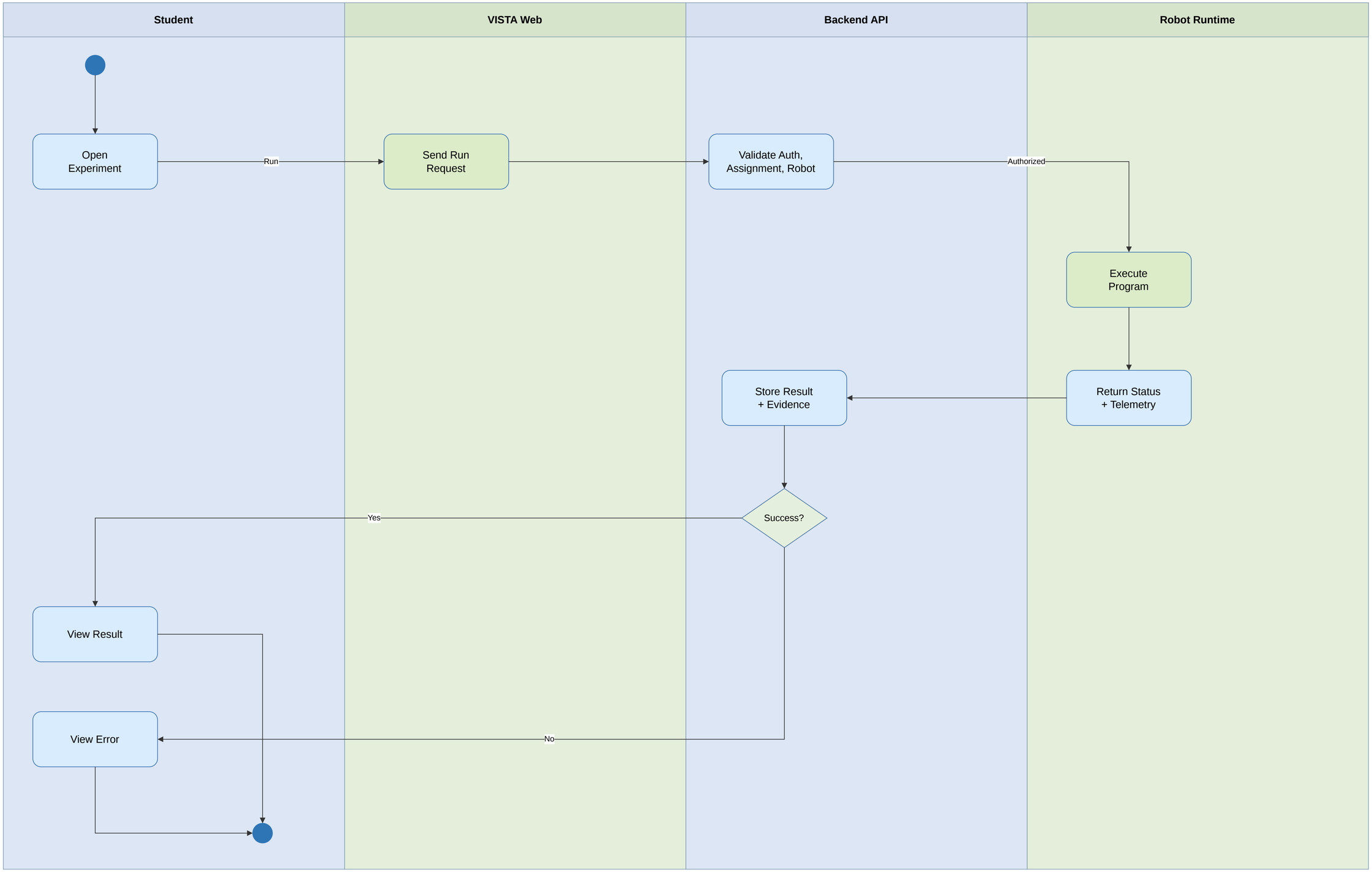
05 Activity WF-02 - Learning & Assignment Management

Activity Diagram



06 Activity WF-03 - Robot Experiment Execution

Activity Diagram



07 Activity WF-04 - AI/Data Integration & Teacher Review

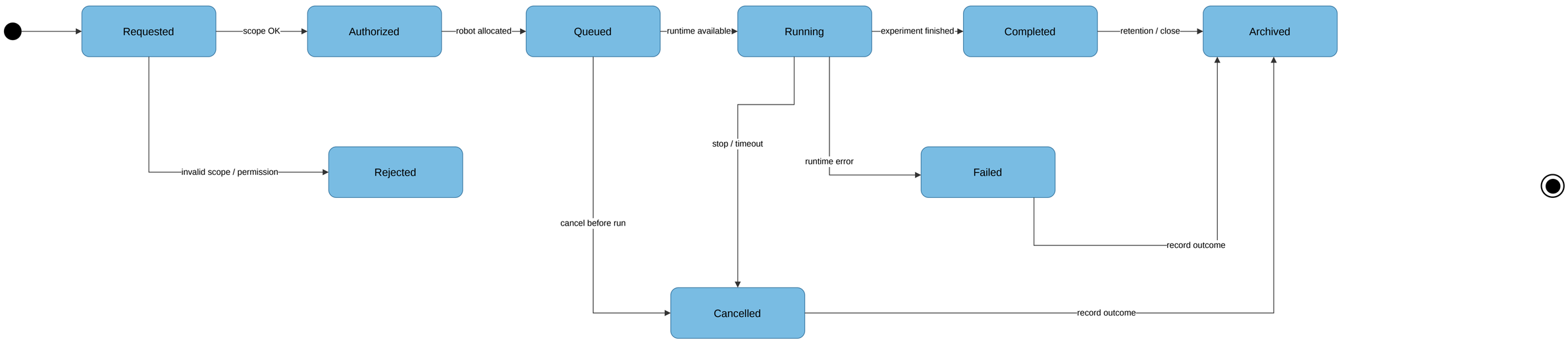
Activity Diagram



State Machine Diagram

08 State Machine - Experiment Session

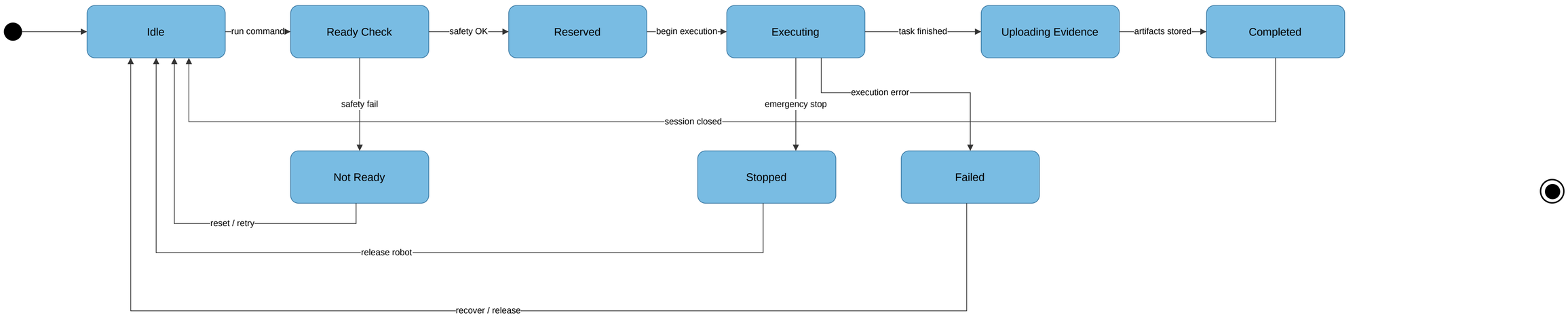
Proposed state machine for control_sessions / learning_experiments; validate transitions against backend.



State Machine Diagram

09 State Machine - Robot Execution

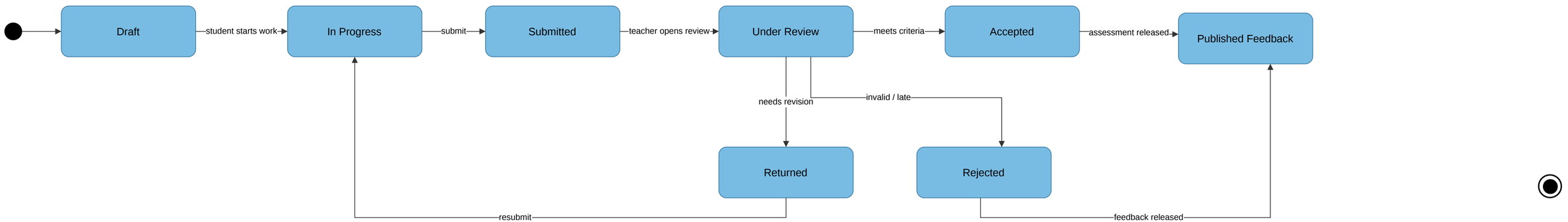
Proposed state machine for runtime execution; validate enum/status names in code.



State Machine Diagram

10 State Machine - Learning Submission

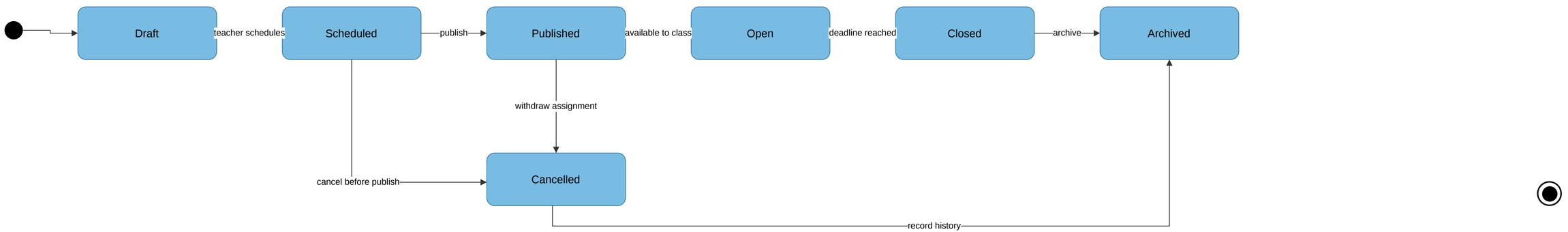
Proposed state machine for learning_submissions / learning_assessments.



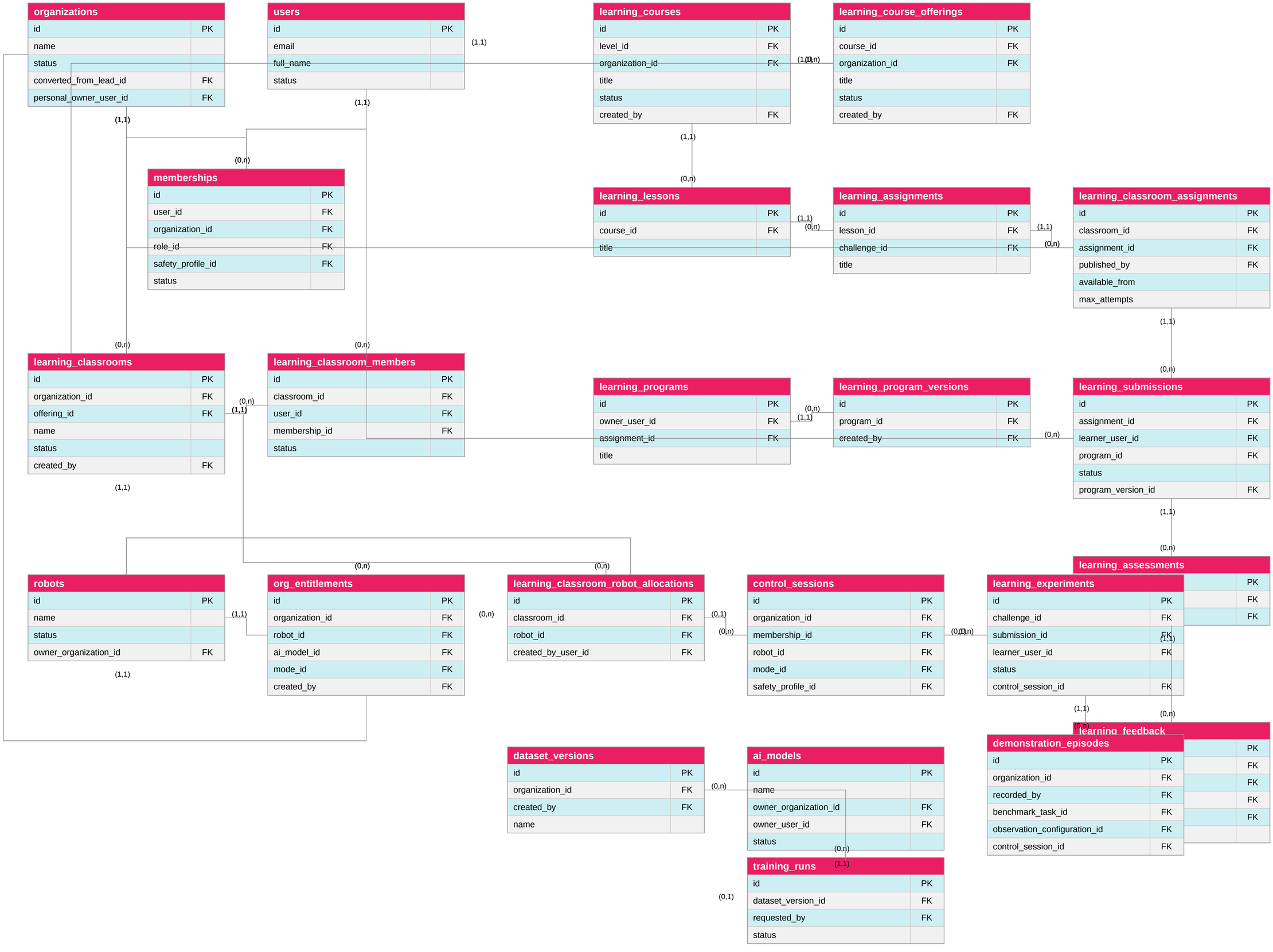
State Machine Diagram

11 State Machine - Classroom Assignment

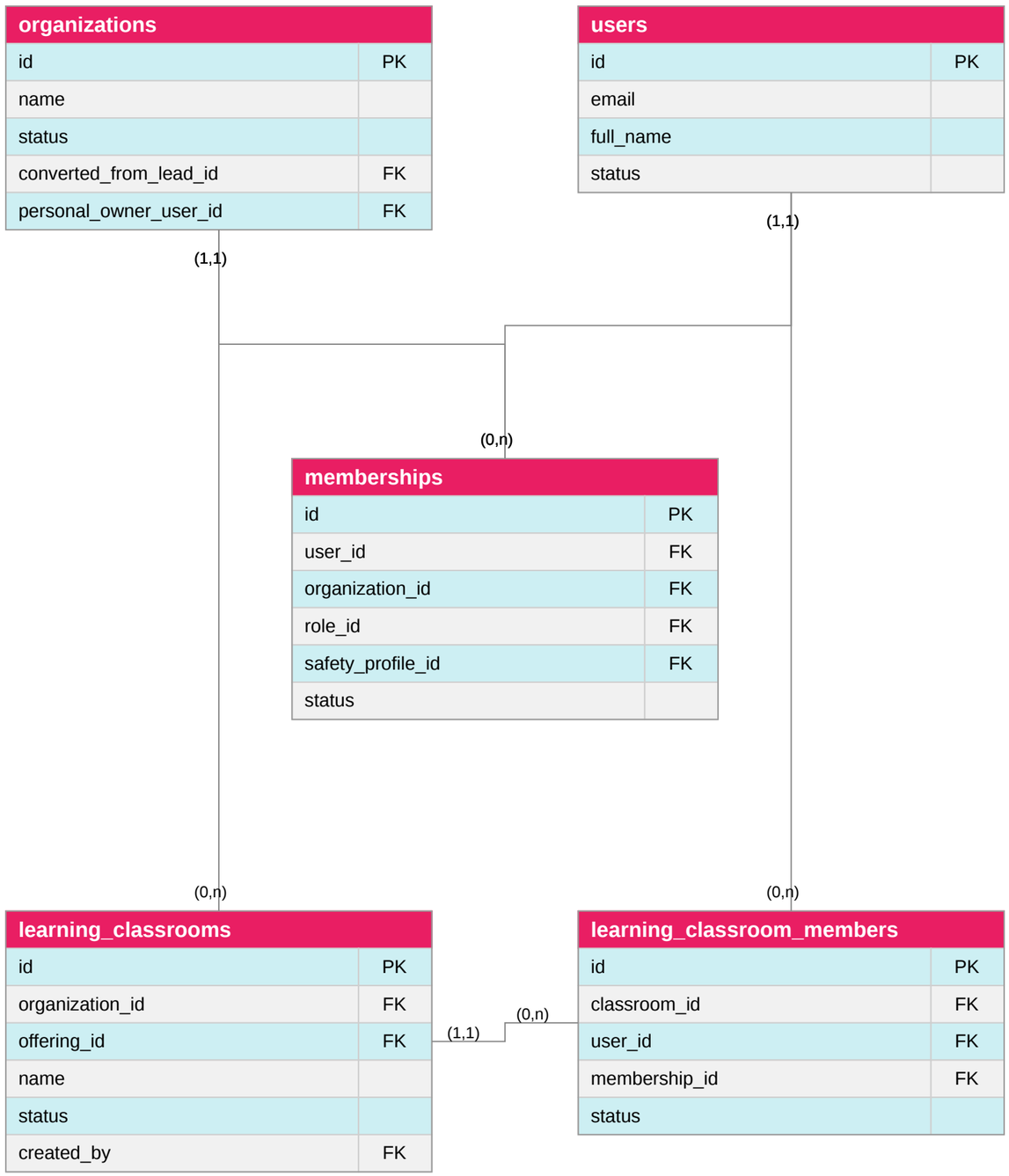
Proposed state machine for learning_classroom_assignments / assignment lifecycle.



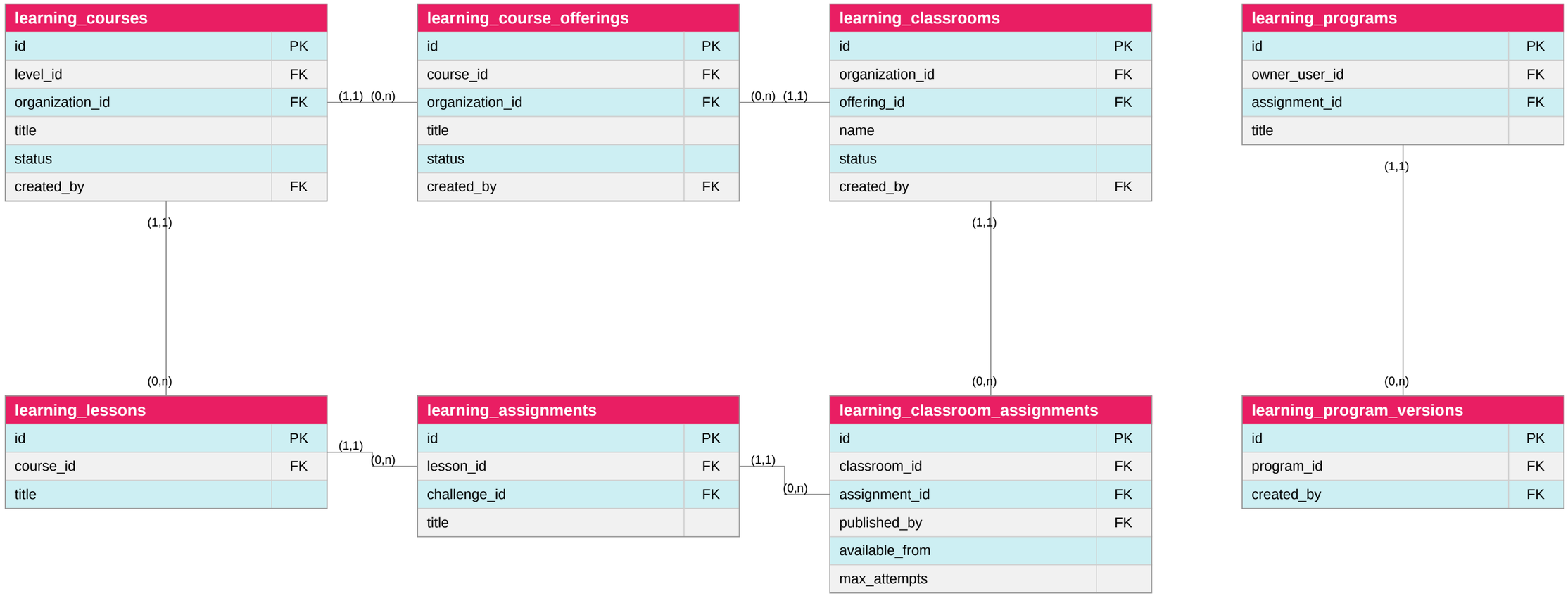
12 Master Logical ERD



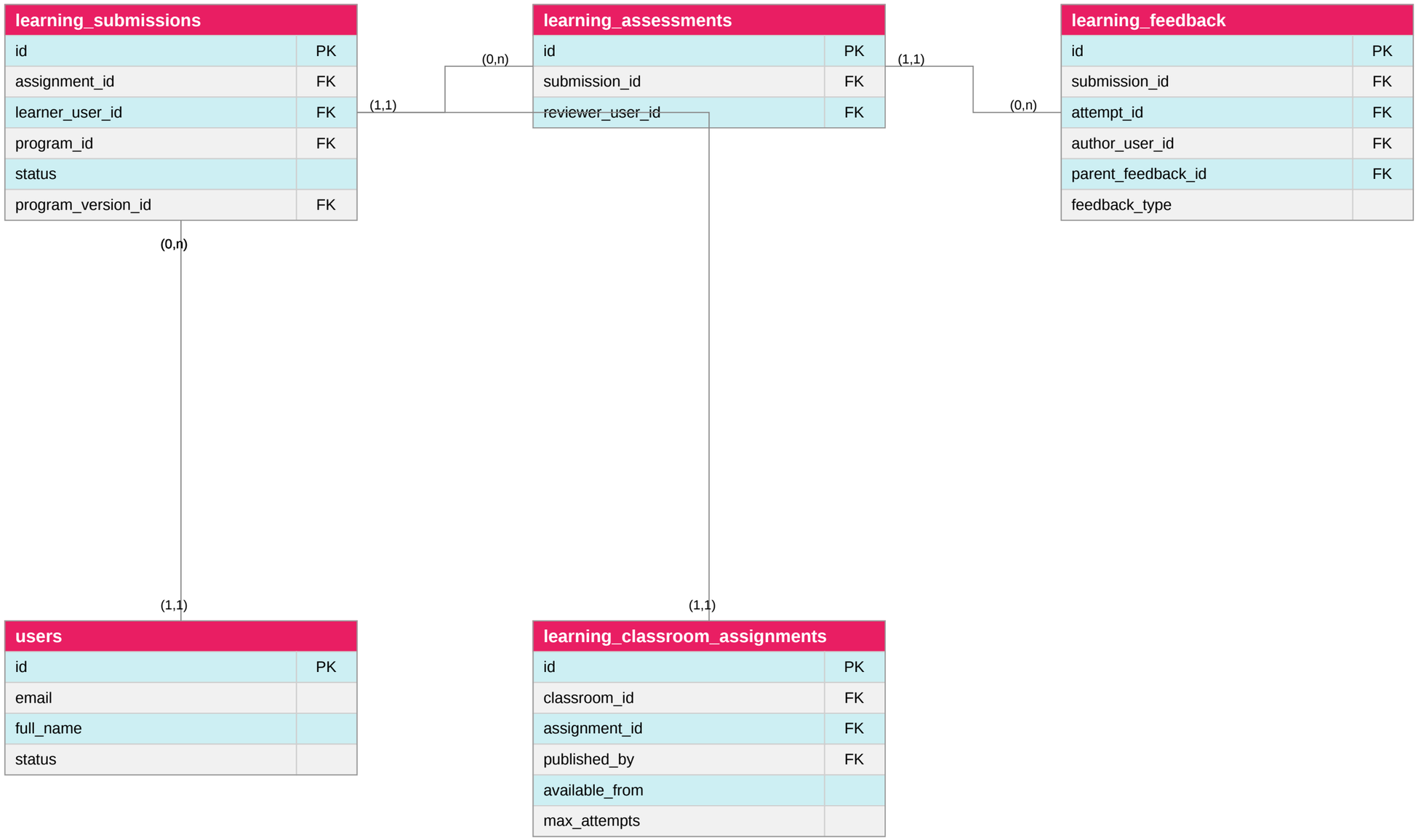
13 ERD - Identity & School



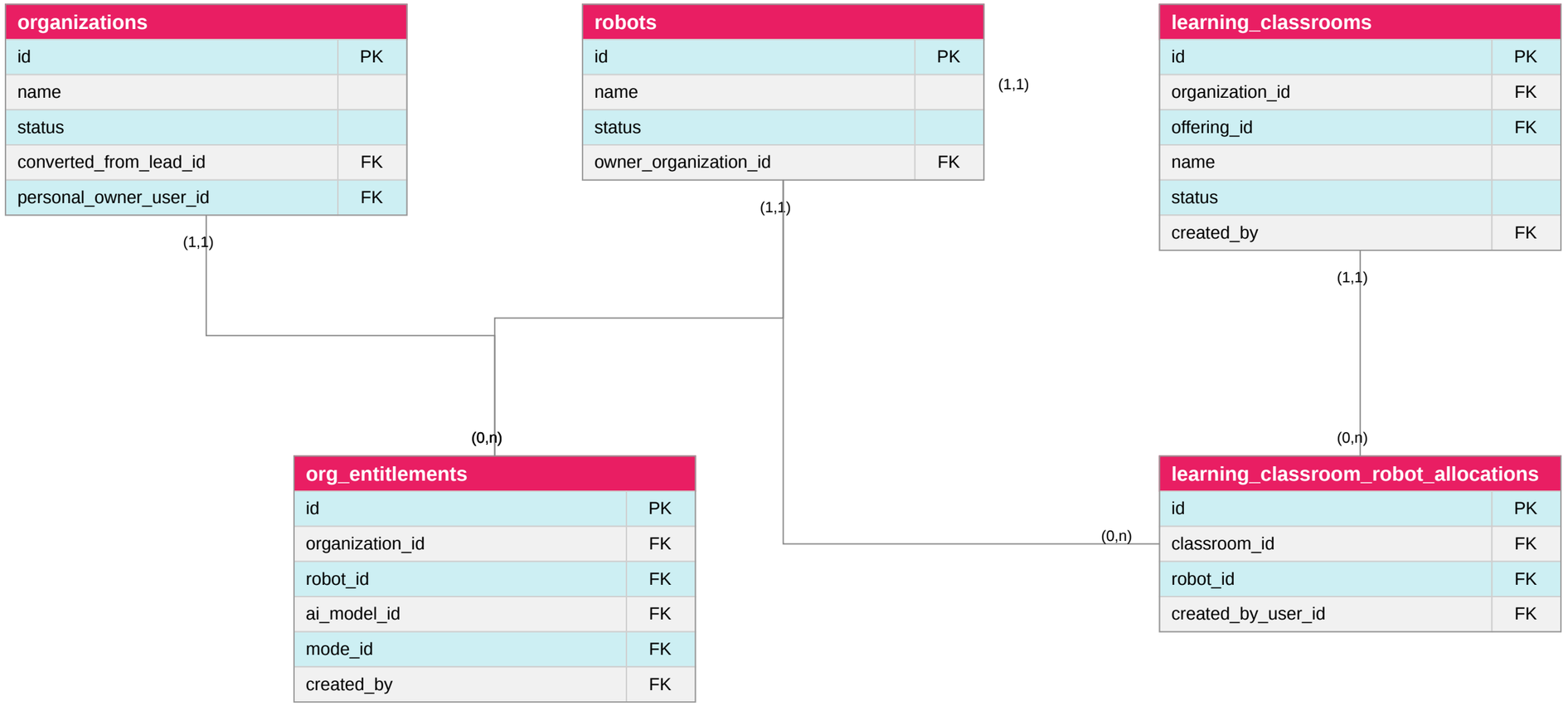
14 ERD - Curriculum & Learning



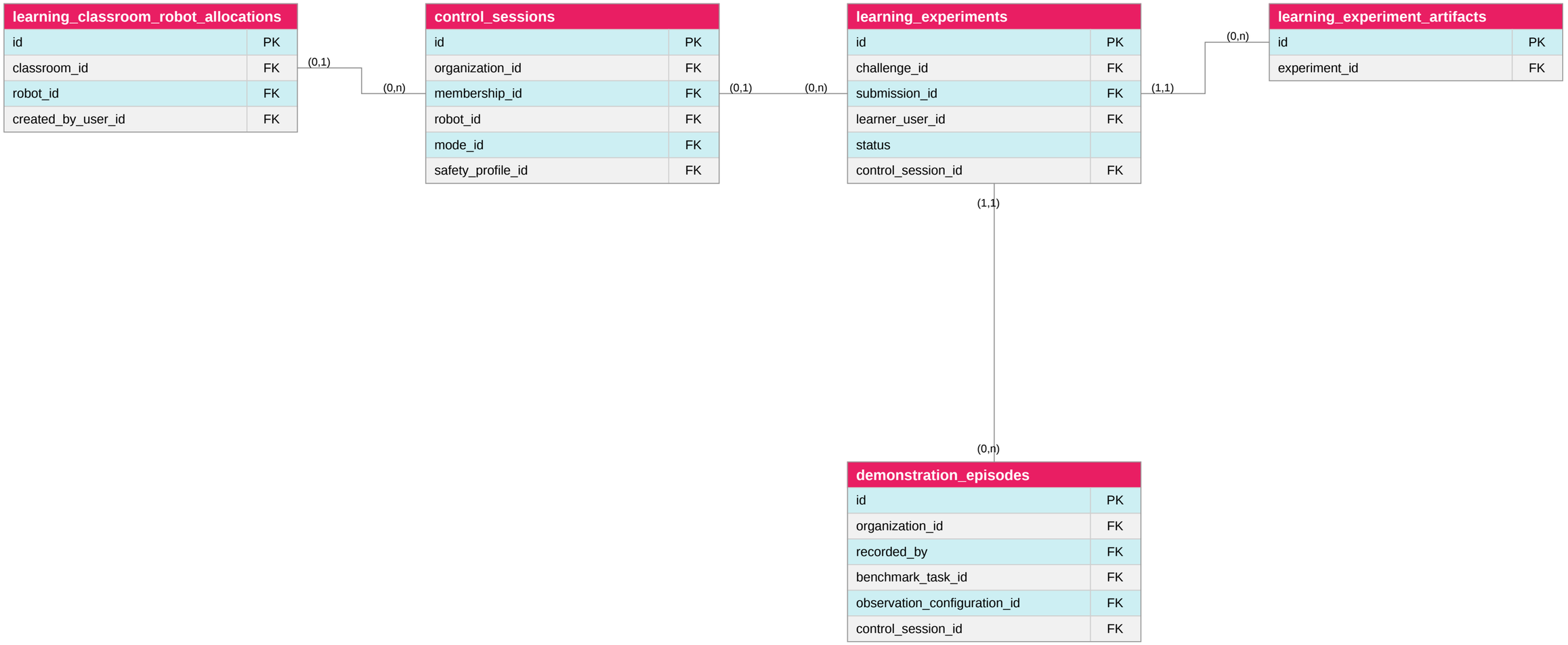
15 ERD - Assessment



16 ERD - Robot Resource Management



17 ERD - Execution & Experiment



18 ERD - AI & Datasets

